

# Python Intermediate Day 4 Checkpoint

Exception Fundamentals

Designing with Exceptions

File System & Serialization

Data with NumPy & Pandas

Mixins

Agentic AI (Development best practice)

Modules (pre-test)

1. Email \*

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2. Name

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3. What happens when an exception is raised inside a try block?

1 point

*Mark only one oval.*

- ☐ A. Program stops immediately
- ☐ B. Python searches for a matching except block
- ☐ C. finally runs first
- ☐ D. else runs automatically

4. The finally block is used to:

1 point

*Mark only one oval.*

- ☐ A. Catch exceptions
- ☐ B. Stop execution
- ☐ C. Execute cleanup code regardless of errors
- ☐ D. Retry execution

5. To create a custom exception, you should inherit from:

1 point

*Mark only one oval.*

- ☐ A. object
- ☐ B. Error
- ☐ C. Exception
- ☐ D. Base

6. Exception chaining preserves:

1 point

*Mark only one oval.*

- ☐ A. Memory usage
- ☐ B. Original error context
- ☐ C. Performance
- ☐ D. Variable scope

7. Logging errors to a file is important because it:

1 point

*Mark only one oval.*

- ☐ A. Makes code faster
- ☐ B. Replaces exceptions
- ☐ C. Keeps persistent error records
- ☐ D. Stops stack traces

8. Which module helps process directories?

1 point

*Mark only one oval.*

- ☐ A. sys
- ☐ B. os
- ☐ C. math
- ☐ D. csv

9. Which modern module simplifies file path handling?

1 point

*Mark only one oval.*

- ☐ A. file
- ☐ B. pathlib
- ☐ C. dirlib
- ☐ D. system

10. JSON serialization converts:

1 point

*Mark only one oval.*

- ☐ A. Python objects to binary
- ☐ B. JSON to HTML
- ☐ C. Python objects to JSON string format
- ☐ D. CSV to dict

11. A custom decorator typically:

1 point

*Mark only one oval.*

- ☐ A. Rewrites the class definition
- ☐ B. Wraps a function to extend behavior
- ☐ C. Deletes attributes
- ☐ D. Prevents inheritance

12. Which file mode opens a file for writing and truncates it?

1 point

*Mark only one oval.*

- ☐ A. "r"
- ☐ B. "a"
- ☐ C. "w"
- ☐ D. "w"

13. Pickle differs from JSON because pickle:

1 point

*Mark only one oval.*

- ☐ A. Is human readable
- ☐ B. Supports only strings
- ☐ C. Can serialize complex Python objects
- ☐ D. Is safer for untrusted input

14. A mixin is best described as:

1 point

*Mark only one oval.*

- ☐ A. A full standalone class
- ☐ B. A reusable feature class that adds methods
- ☐ C. A decorator
- ☐ D. A module

15. Aspect-Oriented Programming (AOP) is primarily about:

1 point

*Mark only one oval.*

- ☐ A. UI design
- ☐ B. Data structures
- ☐ C. Separating cross-cutting concerns
- ☐ D. JSON parsing

16. An Agent AI system typically:

1 point

*Mark only one oval.*

- ☐ A. Only predicts text
- ☐ B. Executes predefined static rules
- ☐ C. Uses tools, memory, and reasoning to complete tasks
- ☐ D. Is the same as a decorator

17. What is scikit-learn primarily used for?

1 point

*Mark only one oval.*

- ☐ A. Web scraping
- ☐ B. Machine learning models
- ☐ C. File serialization
- ☐ D. Decorators

18. Mixins vs decorators difference:

1 point

*Mark only one oval.*

- ☐ A. Mixins wrap functions
- ☐ B. Decorators add inheritance
- ☐ C. Mixins extend classes via inheritance; decorators wrap functions/classes
- ☐ D. They are identical

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